HINTS & SOLUTIONS

Q.1 (B)

No. of geometrical isomers in $[Pt(NO_2)(NH_3)(NH_2OH)(Py)]^+ = 3$ and

No. of stereoisomers for $[Pt(Br)(Cl)(I)(NO_2)(NH_3)(Py)] = 15 + 15$

Q.2 (D)

$n_{COCl_3.6H_2O} = 0.01\,\text{moles}$

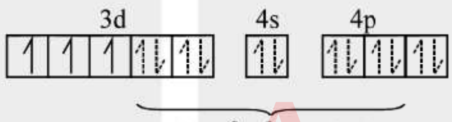
number of moles of ions $= \dfrac{0.6\times10^{22}}{6\times10^{23}} = 0.01$

Complex is = $[Co(H_2O)_4Cl_2]\,Cl.2H_2O$

[0.01 mol of above complex will give 0.01 moles of ions.

Q.3 (B)

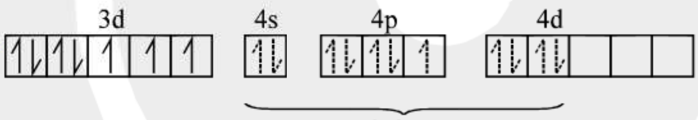
(A) Cr^{3+} in $[CrF_6]^{3-}$



Hyb.: d^2sp^3, octahedral

$\mu_{eff} = 3.89$ B.M.

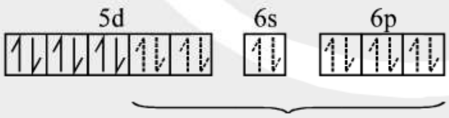
Co^{++} in $[Co(H_2O)_6]^{++}$



Hyb.: sp^3d^2, octahedral

$\mu_{eff} = 3.89$ B.M.

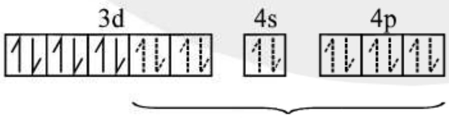
(B) Ir^{3+} in $[IrF_6]^{3-}$



Hyb.: d^2sp^3, octahedral

$\mu_{eff} = 0$ i.e. diamagnetic

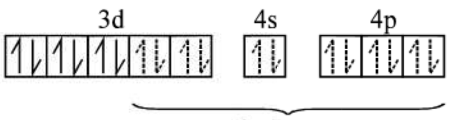
Co^{3+} in $[Co(NH_3)_6]^{3+}$



Hyb.: d^2sp^3, octahedral

$\mu_{eff} = 0$

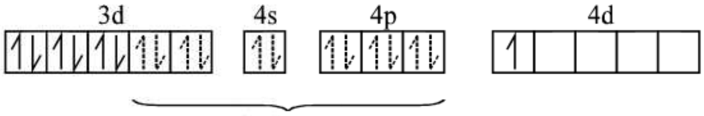
(C) Co^{3+} in $[Co(H_2O)_6]^{3+}$



Hyb.: d^2sp^3, octahedral

$\mu_{eff} = 0$

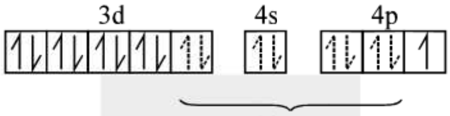
Co^{++} in $[Co(CN)_6]^{4-}$



Hyb.: d^2sp^3, octahedral

$\mu_{eff} = 1.732$ B.M.

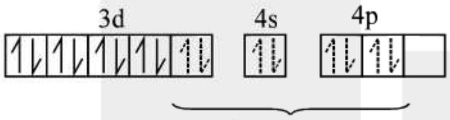
(D) Cu^{++} in $[Cu(NH_3)_4]^{++}$



Hyb.: dsp^2, square planare

$\mu_{eff} = 1.732$ B.M.

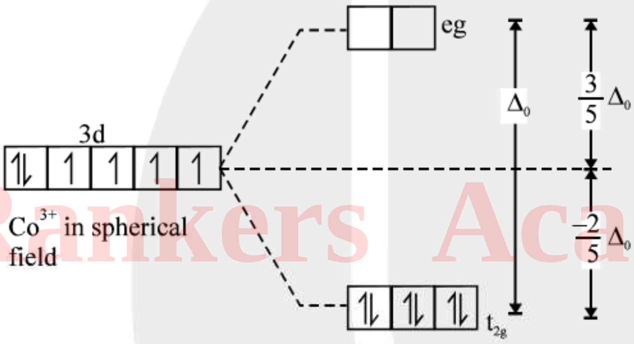
Ni^{++} in $[Ni(NH_3)_4]^{++}$



Hyb.: dsp^2, square planare

$\mu_{eff} = 0$

Q.4 (C)

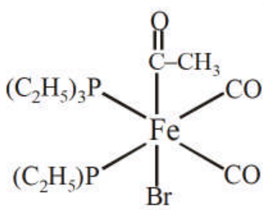


Co^{3+} in $[Co(NH_3)_6]^{3+}$

$P < \Delta_0$

C.F.S.E. $= -6 \times \frac{2}{5}\Delta_0 = -2.4\Delta_0$

Q.5 (A)



$[Fe(CH_3CO)(Pet_3)_2(CO)_2Br]$

Q.6 (C)

$\overset{VIII}{[Os(ONO)(O)_2(O_2)(SCN)(H_2O)]^+ OH^-}$

O_2 will be negative ligand because it is written prior to SCN^-, hence status of O_2 may be either O_2^- or O_2^{2-}, with O_2^- only charge on co-ordination sphere is proved.

Q.7 (A)

If the solution is blue, then absorbed wavelength of light will be of orange to transfer electron from t_{2g} to e_g.

If the solution is green, then absorbed wavelength of light will be of red to transfer electron from t_{2g} to e_g.

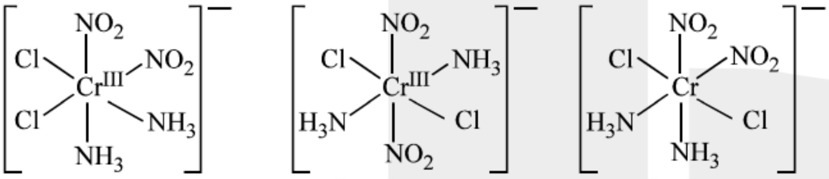
As $\lambda_{orange} < \lambda_{red}$, hence

$\therefore$ CFSE for blue solution will be maximum.

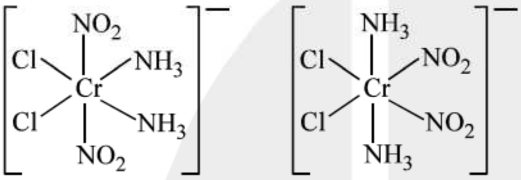
Q.8 (A)

Q.9 (D)

$[CrCl_2(NO_2)_2(NH_3)_2]^-$



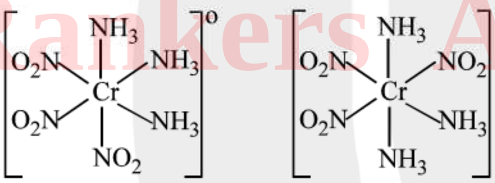
cis   trans   trans



trans   trans

$\Rightarrow$ No. of Geo. Iso = 5

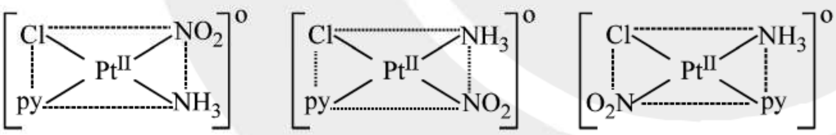
(ii) $[Co(NO_2)_3(NH_3)_3]$



facial or cis form   trans or meridional form

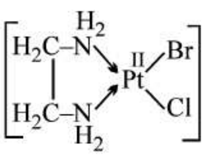
$\Rightarrow$ No. of Geo. Iso. = 2

(iii) $[PtCl(NO_2)(NH_3)(py)]$



No. of Geo. Iso = 3

(iv) $[PtBrCl(en)]$



No. of Geo. Iso = 0

Q.10 (B)

Q.11 (B)

$CoCl_3.4NH_3 + AgNO_3 \longrightarrow AgCl$

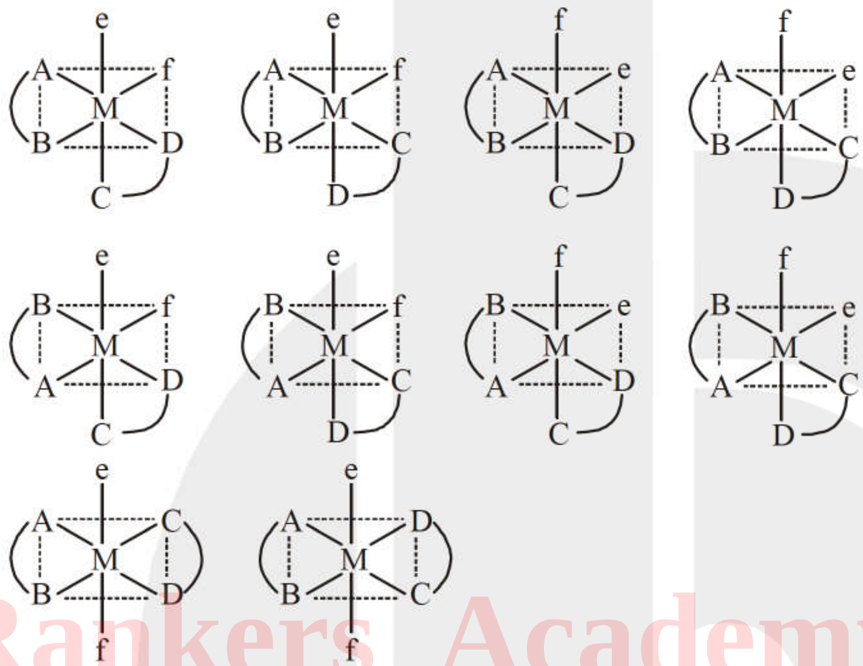
50 ml (0.2 M) $\qquad$ 1.435 gm

$= 0.01$ mole $\qquad$ moles $= \frac{1.435}{143.5} = 0.01$ mole

Thus 1 mole of complex forms 1 mole of AgCl

so $[CoCl_2(NH_3)_4]Cl$

Q.12 (C)



Q.13 (D)

Q.14 (C)

Q.15 (C)

Q.16 (D)

In complexes $[Rh(H_2O)_6]^{3+}$ and $[Co(H_2O)_6]^{3+}$, central metal cations have same oxidation state as well as same ligands and they fall in same group, but Δ_0 of $[Rh(H_2O)_6]^{3+} > \Delta_0$ of $[Co(H_2O)_6]^{3+}$ because Rh^{3+} has high zeff value than Co^{3+}

Q.17 (C)

Q.18 (A)

Q.19 (C)

Q.20 (B)

Q.21 (A)

Q.22 (C)

Q.23 (A

$Mn(CO)_5 + \bar{e} \longrightarrow [Mn(CO)_5]^-$

less stable (O.A.) $\qquad$ More stable, as E.A.N of Mn = 36 (Kr)

Q.24 (A)
If PF_3 is better π acceptor than CO, than C–O bond order in II complex is more than I complex and the C–O bond length in I > II.

Q.25 (D)

Q.26 (C)

Q.27 (B)

Q.28 (A)

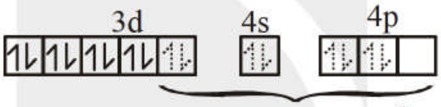
Q.29 (C)
(C) EDTA is hexadentate ligand. When it forms complexe with a metal cation, the hybridization of central atom will be sp^3d^2

Q.30 (C)
Ethylene accepts electron pair from filled d-orbital of Pt^{2+} into its vaccant **antibonding M.O**.

Q.31 (D)

Q.32 (A)

(A) Ni^{2+} $(3d^8)$ in $[Ni(NH_3)_4](NO_3)_2 \cdot 2H_2O$

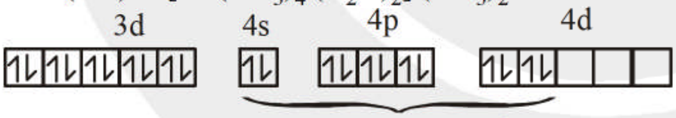


Hybridisation : dsp^2
$\mu_{eff} = 0$

Complex conducts electricity

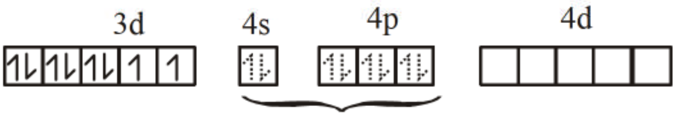
(B) $[Ni(NH_3)_2(H_2O)_2](NO_3)_2 \cdot 2NH_3$
The complex will not exist due to presence of ammoniates i.e. NH_3 as solvent of crystallization because NH_3 cannot lie in voids of crystal lattice in presence of H_2O.

(C) Ni^{2+} $(3d^8)$ in $[Ni(NH_3)_4(H_2O)_2](NO_3)_2$



Hybridisation : sp^3d^2
$\mu_{eff} = 2.8$ B.M.
Complex conducts electricity due to ionic complex.

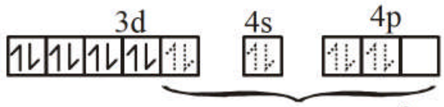
(D) Ni^{2+} $(3d^8)$ in $[Ni(NO_3)_2(H_2O)_2]$ NO_3^- weak ligand



Hybridisation : sp^3
$\mu_{eff} = 2.8$ B.M.
Complex does not conduct electricity due to neutral complex.

Q.33 (C)

(A) Ni^{2+} $(3d^8)$ in $[Ni(NH_3)_4](NO_3)_2 . 2H_2O$



Hybridisation : dsp^2

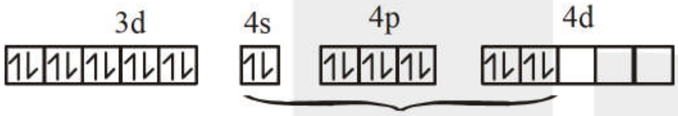
$\mu_{eff} = 0$

Complex conducts electricity

(B) $[Ni(NH_3)_2(H_2O)_2](NO_3)_2 . 2NH_3$

The complex will not exist due to presence of ammoniates i.e. NH_3 as solvent of crystallization because NH_3 cannot lie in voids of crystal lattice in presence of H_2O.

(C) Ni^{2+} $(3d^8)$ in $[Ni(NH_3)_4(H_2O)_2](NO_3)_2$

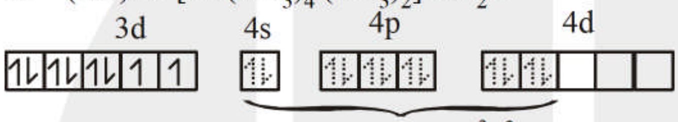


Hybridisation : sp^3d^2

$\mu_{eff} = 2.8$ B.M.

Complex conducts electricity due to ionic complex.

(D) Ni^{2+} $(3d^8)$ in $[Ni(NH_3)_4(NO_3)_2]. 2H_2O$



Hybridisation : sp^3d^2

$\mu_{eff} = 2.8$ B.M.

But complex does not conduct electricity due to neutral complex.

Q.34 (B)

Ni^{2+} $(3d^8)$: For Cordination number = 6, $\mu_{eff} = 2.8$ B.M.

For Cordination number = 4 (square planar), $\mu_{eff} = 0$

Q.35 (B)

Q.36 (C)

Q.37 (A) (B) (C)

Q.38 (B) (C) (D)

O_2 molecules has configuration

$$\sigma 1s^2 \ \overset{*}{\sigma} 1s^2 \ \sigma 2s^2 \ \overset{*}{\sigma} 2s^2 \ \sigma 2p_x^2 \ \begin{matrix} \pi 2p_y^2 & \overset{*}{\pi} 2p_y^1 \\ \pi 2p_z^2 & \overset{*}{\pi} 2p_z^1 \end{matrix} \ \overset{*}{\sigma} 2p_x$$

Thus any electron gained will enter $\overset{*}{\pi} 2p$ and result in decrease of O_2 bond order and thus O_2 bond length increases.

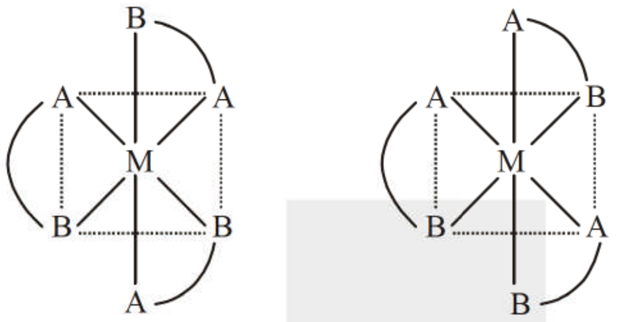
Q.39 (A) (B) (D)

Q.40 (B) (D)

Q.41 (A) (B) (C) (D)

Q.42 (A) P, Q (B) P, R (C) R, S

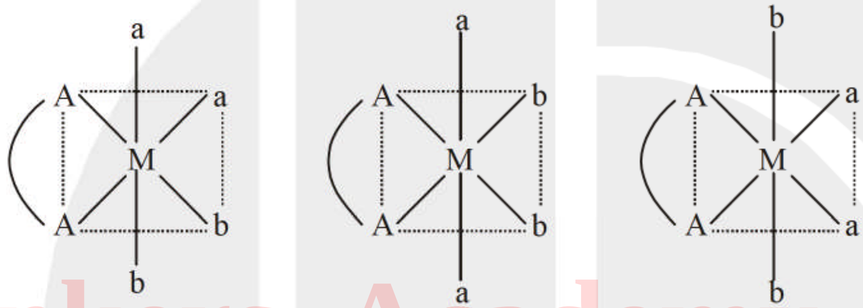
(P)



Cis : optically active Trans : optically active

Number of Geometrical isomer = 2
Number of optical isomer = 4
Total number of space / stereo isomers = 4

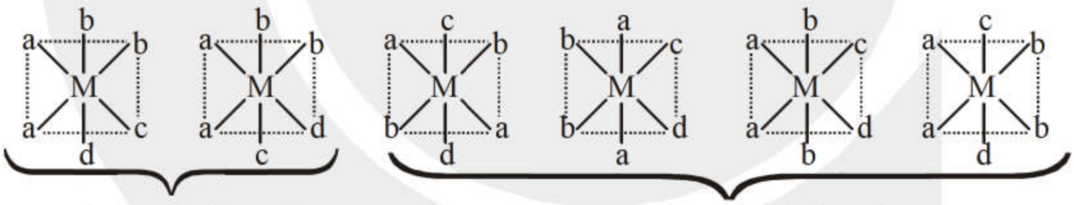
(Q)



Cis : optically active Trans : optical isomer Trans : optical isomer

Number of geometrical isomer = 3
Number of optical isomer = 2
Total number of space isomer = 4

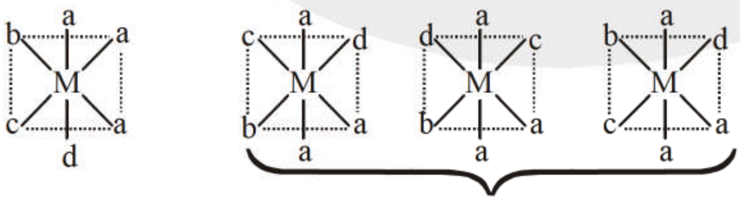
(R)



Cis : optically active Trans : optically inactive

Number of geometrical isomer = 6
Number of optical isomer = 4
Total number of space / stereo isomer = 8

(S)



Cis : optically active Trans : optically inactive

Number of geometrical isomer = 4
Number of optical isomer = 2
Total number of space / stereo isomer = 5

Q.43 (A) S, (B) R, (C) Q, (D) P

(A) $Ma_2b_2c_2\ (\overset{*}{1C} + 4T)$; Optically active = 2

Optically inactive = 4

Total stereoisomers = 6

(B) $Ma_3b_2c\ (1C + 2T)$; Optically active = 0

Optically inactive = 3

Total stereoisomers = 3

(C) $Ma_2bcde\ (\overset{*}{6C} + 3T)$; Optically active = 12

Optically inactive = 3

Total stereoisomers = 15

(D) $M(AB)_3\ (\overset{*}{1C} + \overset{*}{1T})$; Optically active = 4

Optically inactive = 0

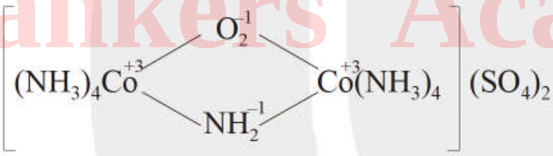
Total stereoisomers = 4

Q.44 0004

If the central metal ion having $t_{2g}^6\ eg^\circ$ configuration than Fe and Co show +2 and +3 oxidation state respectively.

In the presence of O_2^-, complex show paramagentic behavieour

(i) $K_4[Fe(CN)_6]$ $\longrightarrow$ diamagnetic

(ii) 

$\longrightarrow$ paramagentic

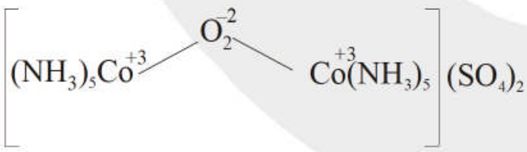
(iii) $K_4[Fe^{+2}(CN)_5O_2^{-1}]$ $\longrightarrow$ paramagentic

(iv) $K_3[Co^{+3}(CN)_5O_2^{-1}]$ $\longrightarrow$ paramagentic

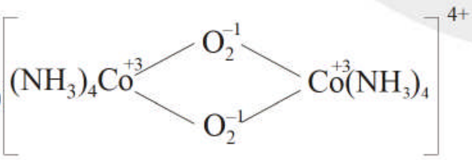
(v) $[Fe^{+2}(CN)_5O_2^{-2}]^{-5}$ $\longrightarrow$ diamagnetic

(vi) $Na_2[Fe^{+2}(CN)_5NO^+]$ $\longrightarrow$ diamagnetic

(vii) $K_4[Co^{+3}(CN)_5O_2^{-2}]$ $\longrightarrow$ diamagnetic

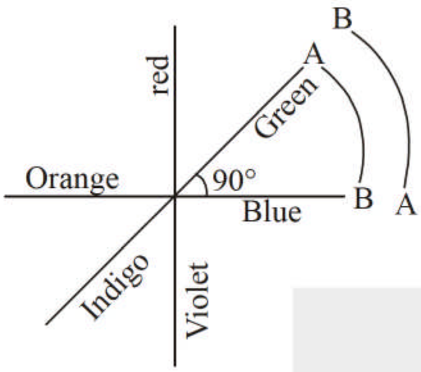
(viii) 

$\longrightarrow$ diamagnetic

(ix) 

$\longrightarrow$ paramagentic

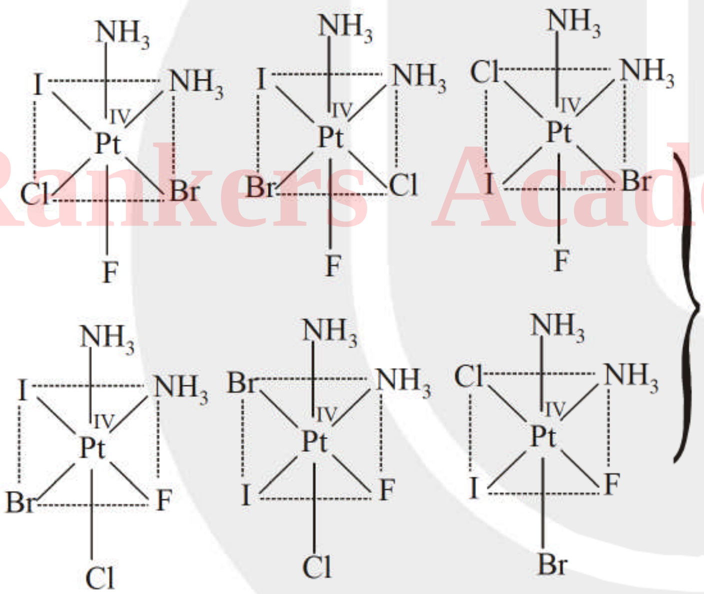
Q.45 24



The adjacent position at 90° (which are different) =12
Hence total possibility = 24

Q.46 0008
Total number of Geometrical isomers in $[Pt(H_2N-CH(CH_3)-COO^-)_2]$ = 4
Total number of Stereo isomers in $[Pt(gly)_3]^+$ = 4
Sum of Geometrical isomers and stereo isomers of given complexes = 8

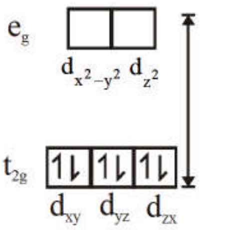
Q.47 12



All cis are optically active

Hence number of optical isomers =12

Q.48 60



Ni^{4+} in $\overset{+IV}{[NiF_6]^{2-}}$